The American Therapist.

Vol. II. New York, January 15th, 1894. No. 7.

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The American Therapist.

A MONTHLY RECORD OF MODERN THERAPEUTICS,

WITH PRACTICAL SUGGESTIONS RELATING TO THE CLINICAL APPLICATIONS OF DRUGS.

VOL. II. NEW YORK, JANUARY 15th, 1894. NO. 7.

Original Articles.

NOTES ON RECENT THERAPEUTICS.

By Oscar H. Merrill, M. D.

Whoever reads the history of Therapeutics will find there records of much faithful work in many directions—records not infrequently of hope deferred. He will find there also a tolerably full account of human credulity, of human weakness and of human cupidity. The same faulty methods of reasoning are followed century after century. *Post hoc ergo propter hoc*, wrecks as many therapeutists to-day as it ever did, notwithstanding its fallacies have been demonstrated so often as to make mention of the subject distressing. It might be expected that half educated physicians, without preliminary, scientific training, would fall into this error; but when some of the brightest men in the profession—men who have presumably travelled the paths of logic and induction all their lives, go the same way, it shows pretty plainly what must be the inherent difficulties of the subject; and that for the proper discussion of therapeutic questions, no caution can be quite great enough and no learning quite profound enough.

The list of dead theories and abandoned remedies grows longer each year, and the experience of the past is as little heeded in the medical as in the financial world.

Acuteness of intellect and extent of education can, it seems, no more keep a man straight in medicine, than they can in religion or politics. Men, who for years have been esteemed well balanced authors and practitioners, become “a little crazy” on some one therapeutic measure and enthusiastically advocate its employment in all sorts of unsuitable cases. Good illustrations of this form of mental activity may be found in the literature of hydrotherapy and of electricity.

Thus, it has been stated that every case of typhoid fever may be made to end in recovery by the proper use of cold baths; and yet this writer knew in how many ways the disease may kill the patient—some of them almost accidental in their nature; he knew that perforation has occurred many days after the disappearance of pyrexia; he knew that in some fatal cases the temperature never exceeded 100°F.

Occasionally such a man after sowing dogmatic statements broad cast for a few years becomes insane enough to be confined in an asylum; sometimes advancing age with its mental deterioration is the evident cause; sometimes these acts are the work of advertising schemers; but generally the explanation is to be found in that mental substratum which permits otherwise sane and well educated persons to entertain monstrous opinions concerning the most ordinary matters.

The best work in Therapeutics is now carried on quietly without brass bands or sensational announcements. A few earnest men in each civilized country are patiently working out the physiological action of drugs, as a basis for a more rational therapy. The significance of much of this work, is not always manifest on superficial examination; but it already forms an important part of our working knowledge, and is gradually crowding out venerable empiricism which has heretofore occupied so prominent a place in medical practice, whether regular or irregular.

Manufacturing Chemists.

No one will deny the value of some of the work done by the manufacturing chemists. Some synthetic compounds have been produced by chemical processes which we should not like to give up, and some improved forms of administering the older remedies are due, at least in part, to their ingenuity. Nevertheless, let any one not accustomed to it read patiently for a few months, the current numbers of half a dozen of what have been called “the minor medical journals,” or let a careful inspection of the advertising pages of the major journals be made, and it will be seen that large classes of medical men—perhaps a majority—have been completely deceived by these shrewd fellows. They have reached a refinement and a delicacy in their commercialism which will compare favorably with the court intrigues of oriental countries. Every prejudice, every weakness, every conceit of “the under medical world” is played upon with consummate skill and with amazing success.

Take, for instance, acetanilid, which, on account for its cheapness has been made to enter into numberless compounds, and every known language is ransacked in the search after new compound names which may be trade-marked. It is not alone the laity that is deceived, but graduates of reputable medical schools are prescribing, and, indeed, dispensing tons of this stuff, and often without knowing its composition.

Animal Extracts.

For thousands of years animal substances of various degrees of nastiness have been used as medicines. In fact some of the darkest chapters of human history are those relating to this subject. Sundry cognate superstitions are extant to this day, even among the nobility of civilized countries. In ancient times weird ceremony and occultism lent their charms to keep up interest in the matter; while in these “scientific times” the influence of a great hyphenated name has rekindled a fire which was merely flickering and which seemed to be in danger of going out altogether.

Since Brown-Sequard made himself young again by his well known treatment of senility great attention has been paid to the subject of animal extracts. From every corner of the earth have come workmen—some of them skilled workmen—to cultivate this promising field. One result of their labor is a mass of literature which contains much that is premature, much that is fantastic, much that is commercial; and it is difficult not to believe that some of it is closely connected with unsoundness of mind.

Another result is the new method of treating myxedema by the administration of thyroid glands—raw, cooked, desiccated, or in the form of extracts. Recently this treatment has been used in several cases with marked success, and already the air is full of rumors. One writer said a few months ago: “The success of this treatment is sure to create a ‘*boom*’ in animal extracts for various diseases.” He proved to be a true prophet, and we are in the midst of this *boom*.

It may perhaps be open to question whether there was need of any more “booms.” We have had “booms” in tuberculin, in coal-tar, in ovariotomy, if not in common sense. Some of them are still with us, though not in very good condition. After the boomers and the seekers after notoriety have done their worst with animal extracts, the final accounting will probably show some increase of positive knowledge in physiology as well as in therapeutics.

Bacteriology.

Bacteriology having now emerged from noisy babyhood into promising youth, it may not be amiss to ask how far therapeutics has been advanced by its discoveries and what the outlook is for the future. It has been alleged by some clinicians that the bacteriologists have been a little dictatorial, and have carried on their propaganda somewhat after the manner of the Salvation army. Be that as it may, the amount of conscientious and unremunerated work which has been devoted to the subject during the past twelve years is probably beyond the power of conception of any one man.

In spite of numerous mistakes, exaggerations and ludicrous attempts to re-organize therapeutics the bacteriologists have made great additions to our knowledge. The question of bacilliary disease is, however, enormously complex, and can not be settled by a few cultures and a few hasty deductions. Points that were supposed to be settled a few years ago are now under discussion again, and with regard to many of the problems it is still impossible to say just where the truth lies. Many thoughtful men have lately been turning their eyes from the microscope toward the bedside, and are asking themselves whether after all the condition of the soil is not fully as important as the germs which grow there, and they are consequently spending less time searching after germicides. One eminent physician predicts that the very language now used will be unintelligible jargon to future generations of germ seekers, and that the whole subject will have to be recast. Whether this be too strong language or not, it is doubtless true that we are, as yet, only on the threshold of this department of science, and that exact truth can be established only by a closer union of clinical medicine with bacteriological studies. The success of modern methods of preventing disease and the comparative failure of antiseptic and germicidal remedies in the treatment of well developed disease show that the human body is more than a test-tube, even though numerous theories have been tested therein to the discomfiture of the testers.

Creosote.

Creosote still refuses to move on into obscurity with the numerous "cures" for consumption which have recently made their exit from the therapeutic stage. On the contrary it is used to a greater extent than ever before, and the testimony as to its value gets stronger each year. It is interesting to note here that a writer who is by many regarded as the leading American authority on therapeutics feels justified in ignoring the whole matter in the last edition of his work recently published. Such is the perversity of the human mind—or at least of his human mind.

In giving creosote, it will be found that most patients take it in the form of sugar-coated pills more readily than in capsules or liquid mixtures. This statement is deliberately made after thoroughly trying every known method of administration in a large number of cases during a period of eight years.

Patients who have sensitive stomachs can frequently take, at first, only one or two minims per day; but by slowly and carefully increasing the dose, they eventually consume twelve to fifteen minims each day without suffering from gastric irritation.

The success of the creosote treatment is seen most plainly in those patients who have taken it uninterruptedly and in full doses for several years. Many of these people improve in health from year to year, without, however, losing all their symptoms.

Coal Tar.

It has sometimes been claimed by metropolitan physicians that the rural brethren are slow to avail themselves of the various discoveries in medicine and surgery, and that they go on very much as their grand mothers did. Here as elsewhere in the universe there are compensations. Your country doctor can now look back a little, and with regard to many of the startling advances of these latter years he can see that the "advance" has been backward; and he is not quite sorry that his bucolic inertia has kept him from doing urban oöphorectomy upon all his hysterical female acquaintances.

On other occasions he accepts the dicta of the medical centers with alacrity, and refuses to be called off when the fashion changes. This was seen in the case of coal tar. No sooner had the chemists, private docents, assistant physicians, and royal and imperial professors of Germany and Austria, announced the mighty powers of antipyrine, than he began to employ it, and a little later its congeners, to drive fever and pain out of the world.

In many a remote country village this policy is still followed so vigorously that fever patients are kept blue and sometimes black by frequent and heroic doses of coal tar; and yet the great majority survive in spite of the disease and the antipyretic.

Very lately a rampant enemy of coal tar wrote: “While in the medical centers the coal tar antipyretics are being reluctantly abandoned, it will be long ere less enlightened rural practitioner will let this comforting drug slip from his fond grasp.”

Here then we have the two extremes; and here again sound practice lies about midway between them.

These antipyretics when used with skill and caution can be made, in many cases, to replace with advantage, morphine on the one hand, and cold baths on the other. Surely, medicines capable of playing such a part in therapeutics, deserve careful consideration.

245 Prospect Ave., Mount Vernon, N. Y.

Effects of Morphine on the Female Organs.—In a paper recently read before the Obstetric Society of St. Petersburg, Passower related the history of two cases, which confirms the opinion already supported by the observation of others, that the long continued use of morphine eventually leads to atrophy of the female generative organs. In both cases amenorrhœa was present; intra-uterine measurements taken during a period of two years showed a diminution in the size of its cavity from 5.1 to 1.9 inches.—*Exchange.*

ACTIONS OF DRUGS ON THE INTESTINES.

By W. C. Caldwell, M.D.,

Professor of Materia Medica, and Director Pharmacological Laboratory,
College of Physicians and Surgeons, Chicago.

Concluded from page 164.

Methods of Experiment to Determine where a Drug Acts to Produce Catharsis.

(*D*) *By introducing a rubber balloon, to which is attached a graduated rubber tube, through a gastric fistula into the small intestine, and measuring rate of descent and force carrying it onward to anus.*

Technique of the experiment.—Dogs are suited for this. Hess in his experiments used the animal immediately after making the fistula; Brandl and Tappeiner waited for the fistula to heal, and used the same dog a number of times, usually waiting two weeks before using him again, so that he would completely recover from the previous experiment. The balloon is introduced into small intestine and then moderately distended with water through the tube. After it has passed some distance below the pylorus, the cathartic is injected through the tube, the instrument being so constructed that the medicine passes into the intestine just above the balloon; ferrocyanide of potassium is added to the solution so as to tell by its reaction with chloride of iron whether the cathartic solution is in the feces. This method causes no shock and is superior to opening the abdomen to determine peristalsis.

By this method can be learned:

(1) *The location of the peristalsis.* This is done by examining the graduated tube and noticing how many feet been drawn in. If after the cathartic has been introduced above the balloon it does not increase the rate of descent till it reaches the colon, it shows that the drug acts on the colon instead of the small intestine; aloes is an example of this. If it increases its

descent in the small intestine, it shows that the drug acts on the small intestine. If the drug greatly increases its speed in both small and large intestines, then the drug stimulates peristalsis through the whole length.

(2) *The rate of the peristalsis.* This is learned simply by noticing how fast the graduated tube is drawn in.

(3) *The time of peristalsis.* When the tube is not being drawn in there is no peristalsis, at least where the balloon is; when the drug acts on the small intestine the peristalsis occurs early; and when on the large intestine, late.

(4) *The force of the peristalsis.* This can be measured by pulley and bag of shot attached to the graduated tube. Of course it is necessary to learn first the lifting power of the peristalsis in the same dog without a cathartic.

(5) *By injecting a solution of the cathartic into a ligatured loop of intestine in the living animal.* It is better to use a rabbit or dog that has been starved for several days so that the intestine will be empty. After several hours remove the loop and measure the amount of liquid. If it is increased the drug stimulates secretion. It must not be forgotten that the irritation of the ligature stimulates secretion, so it is better to have a similar loop for comparison.

(6) *By injecting an equal quantity of water in two similar loops of intestine in the living animal, and then injecting the cathartic into one of them.* Use a starved animal. After several hours remove the loops and measure the quantity of water in each, if the one containing the cathartic has not diminished as much as the other, then it diminishes absorption.

We have now learned how to determine when there is increased peristalsis, and where it is; when there is increased secretion, and when there is diminished absorption. We have next to try to learn the exact manner in which these are produced, but this is far more difficult:

(*a*) Because the nervous mechanism of the intestines is very complicated, and at present not very much of its physiology is known.

(*b*) Because instead of the nerves with different functions having a separate course, they are all in the same sheaths, so that one kind of fibres at a time cannot be cut. The vagus contains motor, inhibitory motor, sensory, etc., fibres, so that when one is cut all must be cut. The same is true of the splanchnic.

(*c*) Because if a drug appears in the urine, sweat, or milk, it is no proof that it acts in the circulation; for it may act locally, then afterwards be absorbed, but have no action on the intestinal mechanism.

(*d*) Because if a drug cannot be found in the urine, sweat, or milk, it is not proof that it does not act in the circulation, and then afterwards be entirely excreted by the intestinal mucous membrane and pass out with the feces.

(*e*) Because when a drug, given hypodermatically, purges, it is not proof that the drug acts as a cathartic in the circulation, and that it does not act locally, for it may be inactive till it is excreted by the intestinal mucous membrane, and then act locally.

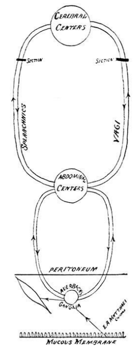


FIG. 5.

At present I will speak of peristalsis only. Perhaps as good a way as any to proceed, though unsatisfactory, is, first, to exclude the brain, and then, second, the abdominal ganglia, and then, third, experiment on the excised living intestine.

(1) Exclude the cerebral centres by cutting the vagi and splanchnics. This is shown in the schematic drawing Fig. 5, in which, for simplicity, one circle represents all the centres in that location. It must not be forgotten that cutting splanchnics affects the intestinal vessels. It is evident that after cutting the vagi and splanchnics the cerebral centres can have no influence on the intestines. After cutting these nerves, give the cathartic:

(*a*) If it does not act as a cathartic, this shows that it acts upon the intestines entirely through the cerebral centres, which now have no influence upon the intestines. This action is probably direct and not reflex.

(*b*) If it acts as a cathartic equally as efficient as it does when the nerves are intact, this shows that it does not act as a cathartic through the cerebral centres. The action is either on some of the abdominal ganglia, or on the local intestinal mechanism.

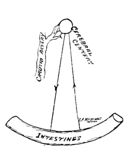


FIG. 6.

(*c*) If it acts as a cathartic, but much weaker than when the nerves are intact, this shows that it acts both upon the cerebral centres and also on some part of the remaining mechanism. This action on the cerebral centers may be direct, or it may be reflex from irritation of the intestinal mucous membrane. So far as I have read, no experiments have been made to determine when the action is direct and when reflex. It seems probable, though, that something might be learned about it in the following way, which is illustrated by Fig. 6. Select two similar starved animals; open abdomens of both and expose one carotid artery in one of them. Inject a solution of the cathartic in the exposed carotid and an equal solution in the exposed intestinal loop of the other animal. Place animals in salt solution in tin vat so as to observe through incision the intestines.

(1) If the drug acts reflexly by irritation of the intestinal mucous membrane, it will act quicker in the animal in which the drug was injected into the intestinal canal (see Fig. 6), because it is there to begin action at once, while in the other it must first be excreted into the intestinal canal. The intestine can be observed easily in the salt solution.

(2) If the drug acts directly upon the cerebral centres, it will act quicker in the animal in which it was injected into carotid artery, because in the other animal it has to be absorbed and diffused through the general circulation before it reaches the centre. Clamping the aorta would make this more accurate were it not that the intestines are so easily disturbed by changing the circulation.

(3) Exclude the abdominal ganglia. This can be done either by extirpating the ganglia, or by using a piece of excised living intestine. (Fig. 7 represents the abdominal ganglia concerned destroyed.) After extirpating the ganglia, and performing tracheotomy, and injecting the cathartic into the small or large intestine, place the animal in a physiological salt solution in tin vat, and observe the action of the drug. This experiment is not very satisfactory, because the vessels are greatly dilated.

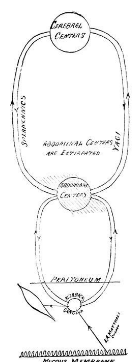


FIG. 7.

(*a*) If the drug does not excite peristalsis, this shows that it acts through the abdominal or cerebral centres. This action may be direct or reflex.

(*b*) If the drug excites equally as strong peristalsis as are produced when the ganglia are intact, this shows that its action is not on the abdominal ganglia but on the intestine.

(*c*) If the drug excites weaker peristalsis, this shows that it acts at both places. This may be direct or reflex.

If it is decided that the drug acts on some part of the intestine to cause the peristalsis, two methods may be used to determine what part it acts upon:

1st Method.—After performing tracheotomy, opening abdomen and destroying abdominal ganglia, place the animal in a tin vat containing warm physiological solution. The first thing is to find out:—

(1) Whether the cathartic acts reflexly by the irritation of the mucous membrane being transmitted to Auerbach's plexus, and from there to the muscles, or

(2) Whether it acts upon some structure in the intestinal wall after being absorbed.

Demonstration.—If it acts reflexly, it will act quicker when injected into intestinal canal than when injected into mesenteric artery. If it does not act reflexly, it will act quicker when injected into the artery.

Next determine where in the intestinal wall the cathartic acts. The points are:

1st. Stimulation of the muscle.

2d. Depression of the inhibitory motor ganglia.

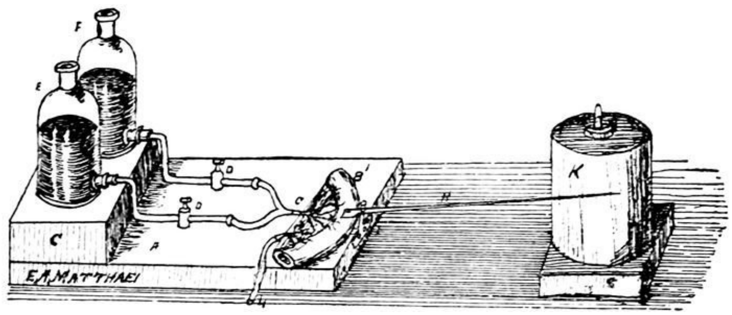
3d. Stimulation of Auerbach's ganglia, and,

4th. Stimulation of the motor fibres.

Demonstration.—If the drug in large dose causes tetanic spasm of the intestinal muscles, which is not affected by electrical stimulation of the inhibitory motor nerves, this shows that the drug acts directly on the muscles. In this way it can be shown how physostigmine acts.

If the drug in large dose increases peristalsis, and electrical stimulation of inhibitory motor nerves has no effect upon it, this shows that the drug paralyzes the inhibitory motor nerves. It can be shown that morphine acts in this way. In fifteen or twenty minutes after giving a dog a grain of morphine hypodermatically, it usually has one movement of the bowels. A small dose of morphine constipates, because it stimulates the inhibitory motor nerves. If the drug increases peristalsis and stimulation of the inhibitory motor nerves lessens the peristalsis, this shows that the drug either stimulates Auerbach's ganglia, or the ends of the motor nerves. By the method above described Jacobi has made some very elaborate experiments with morphine, atropine, muscarine, and physostigmine.

2d Method.—A piece of excised living intestine is placed in a suitable apparatus to keep it moist and warm, and artificial circulation is established by a canula in the artery, connected with two flasks, as seen in the drawing copied from Brunton (Fig. 8), in one of which is pure blood and in the other the poisoned blood, arranged so that either can be turned on at will. The peristalsis is recorded by the lever, shown in the drawing. In this way Ludwig and Salvioli studied the action of a number of drugs.



From Brunton.

Actions of Cathartics in Disease.

Cathartics are not only of use to remove feces, but owing to the physiological relation of the intestinal tract to the other organs, they are often of great service in a number of different pathological conditions. With a knowledge of their mode of action we can select on a rational basis the proper cathartic for the different indications.

Cathartics are indicated:

(1) *To remove dropsical effusions.* In disease of the liver, heart, kidneys, etc., there are sometimes accumulations of serum in the serous cavities and subcutaneous cellular tissue. For this purpose those which stimulate secretion and cause a profuse watery discharge are best suited. When the effusion is due to obstruction in the liver so that the serum leaks from the congested postal vessels, they are most prompt in action, because they drain the water from the postal vessels and lower the high portal pressure. Magnesium sulphate, elaterium, and gamboge are efficient here.

(2) *To remove urea, etc., from the blood.* Sometimes the kidneys are so diseased that they cannot excrete the waste matter from the body, and its

action, when abnormally accumulated, causes serious symptoms. For this purpose use those which cause watery stools, so as to wash the waste out through this channel. If the case is urgent as in uremic convulsions or coma, croton oil is the best to use, because it can be dropped on the tongue and does not have to be swallowed, and, besides, acts in an hour or two. As it is a violent irritant, it should not be continued.

(3) *To lower high systemic pressure.* High arterial tension often increases the disease as sometimes occurs in cerebral hemorrhage, in meningitis, and at the beginning of many acute diseases. Those drugs are of use which cause great dilatation of the intestinal vessels, draining the blood from the other organs, or which cause a profuse watery discharge from the blood. In this way croton oil is of use in cerebral hemorrhage, calomel in meningitis, etc.

(4) *To depress the liver when it is excessively active.* In that condition known as "biliousness," where there is congestion of the liver, and enormous quantities of bile are poured into the intestine, chologogue cathartics are needed, because they remove all this bile from the duodenum, and give the over-active liver rest.

(5) *To deplete the mucous membrane in gastro-duodenitis.* When persons of a certain predisposition over-eat there is congestion or even inflammation of the gastro-intestinal mucous membrane. This extends up the bile-duct, and the swelling occludes it, so that the outflow of bile is obstructed, and soon it diffuses into the blood, and the person becomes jaundiced. The salines, specially sodium salts, are here of use, because they drain away the water without stimulating the mucous membrane, and hence lessen the hyperemia.

(6) *To stimulate the torpid liver.* This is not in biliousness, but when the liver does not secrete bile enough; the patient is constipated, and the stools are too light from lack of bile.

(7) *To remove pathogenetic material from the bowels.* Diarrhea is usually caused by the irritation of toxines formed by the growth of bacteria in the intestinal liquids. In the adult when this is an acute attack, salines are best to wash them out. In the child, when this continues for days, it is better to select those cathartics which are bactericides, so as to kill them, such as calomel and gray powder.

(8) *To remove feces.* This is probably the most common use of cathartics. Those which slightly increase peristalsis and secretion are adapted for this. The drastic purgatives are too strong for this purpose. Among those commonly used are magnesia, sulphur, castor oil, rhubarb, senna, cascara sagrada, jalap, podophyllin, and salines.

(9) *To relieve chronic constipation.* In selecting a cathartic for this purpose we will be guided so far as possible by local conditions. The trouble may be in the small intestine, colon, or rectum. We must decide whether it is due to diminished peristalsis, or secretion, or too rapid absorption. If it is due to a relaxed condition of the muscles of descending colon and rectum, aloes is efficient. If it is due to general relaxation of the intestinal muscles, physostigmine or strychnine may be of use. If there is a diminished secretion or increased absorption the ingestion of more water and foods containing water, as fruits and vegetables, will sometimes be of service.

(10) *To purge the nursing child.*

168 S. Halstead St., Chicago.

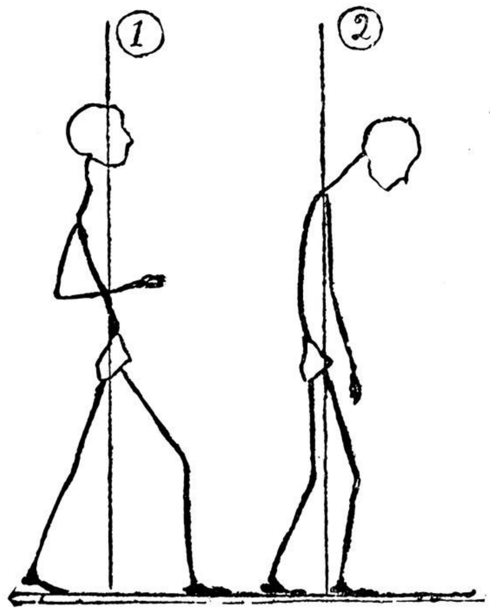
UNREGARDED CAUSES OF ILL-HEALTH IN AMERICAN WOMEN.

By JOHN FORD BARBOUR, M. D.

That ill-health is more common amongst American women of the middle and higher classes than amongst the women of other nations is proven by a great many considerations. In the first place, we have the testimony of many eminent physicians on this point. Dr. Austin Flint, Sr., is said to have declared that if things went on as they are now going, in fifty years it would be well-nigh impossible to find a healthy woman of American descent. Numerous articles have appeared in the medical magazines by such writers as Dr. Mary Putnam-Jacobi, Dr. Engelmann, and many others, calling attention to the alarming and increasing prevalence of ill-health among American women. We have in addition the testimony of such close and careful observers as our novelists, Howells and James, which is of even greater value, as coming from laymen, who would naturally not be so quick to notice such things as physicians. Howells speaks of the "typical American girl, never very sick and never very well." Do we not all know her? And again and again he speaks of her lack of physical development as compared with her European sisters. James, in one of his stories, describes a little girl who comes dashing into the hotel parlor on roller-skates, crying, "Get out of the way." One can see all too plainly from his description, the typical American little girl, with her weak ankles, her thin, flat chest, her feeble little arms and legs, and her lack of proper parental control. Again, in "*A Bundle of Letters*," he makes one of his male heroines say: "The types here, excepting myself, are exclusively feminine. We are thin, my dear Harvard; we are pale, we are sharp. There is something meagre about us; our line is wanting in roundness, our composition in richness. The American temperament is represented by two young girls. These young girls are rather curious types. They are cold, slim, sexless; the physique is not generous, not abundant; it is only the drapery that is abundant."

The bearing of two or three feeble children ought not to make a physical wreck of a woman; but how often do we see this the case? Compare the American mother—with her sick headache, her general physical inefficiency, her everlasting doctors' and druggists' bills, and her two or three delicate, bottle-fed children, with the stout, active German or English matron, and her sturdy brood of eight or ten, every one of whom she has nursed at her own breast.

Let the reader take his stand at some fashionable street-corner on a sunny afternoon and notice carefully the women who pass by. Below are given two figures after Lauder Brunton. The first represents the posture of health; the second, the posture of physical degeneracy.



He will find that about seven out of ten of the women who pass him will present the second posture. In addition to this, they will show by their sallow complexions, thin, flat chests, angular figures, and miserable gait, all the evidences of present or impending ill-health. He will hardly find one woman out of ten with bright eyes, a clear complexion, an erect carriage and a firm step.

It is simply impossible for human beings to live as most American women live, and have good health. After a girl puts on long dresses, she is taught that almost any sort of active bodily movement is unladylike. Her frail little body is enveloped in corsets, which effectually prevents the full development of the important trunk-muscles. At school she gets, perhaps, a little make-belief of calisthenics. During her life in society she waltzes with great ardor, it is true, and it is said that to go through all the evolutions of a

German is equivalent to a walk of fifteen miles; but no one has ever seriously claimed that this form of exercise is conducive to health. After marrying, the American woman reduces physical exertion to the minimum. Any form of physical labor is regarded by her as menial. If the weather is bad, she will not put her foot out of the house for whole days at a time. If she does go out, often she will take a car to ride two squares.

American women take more drugs than other women. The drug-store is ubiquitous with us. They show increased susceptibility to stimulants and narcotics, sensitiveness of the digestion, increase of near-sightedness and weakness of the eyes, early and rapid decay of the teeth, sensitiveness to heat and cold, tendency to nervous exhaustion etc.

Gynecology and abdominal surgery are American contributions to medical science.

This state of affairs is not limited to any particular section of the United States, but extends from Maine to California, and from Dan even unto Beersheba. It is national in its scope. The fact is that ill-health is so common in our women that we have grown accustomed to it, and have, quite unconsciously, set up a low standard of feminine health. If a woman is habitually constipated, dyspeptic, so weak that she can hardly walk a mile, has one or two headaches every month, irregular and painful menstruation, backache, napeache, cold hands and feet, leucorrhea, capricious appetite, and a moderate degree of nervousness and insomnia, she is considered to possess average good health. If she is free from all these, it is a rare exception, a fortunate accident, not a condition of health to be attained by the exercise of reason and common sense.

What, then, are the causes for ill-health which are peculiar to American women? These may be divided into two classes, *viz.*: The unavoidable and the avoidable. Let us briefly survey the unavoidable causes.

By far the most potent of these is our climate, which differs from that of Europe in three respects; it is more variable; the extremes of heat and cold are greater; and the atmosphere is drier.

The reason for the extreme variability of our climate is found in the fact that our mountain chains run north and south, while those of Europe run east and west. In consequence of this, when up in Manitoba Territory King Aeolus reverses his spear and smites upon the side of the mountain, rude Boreas comes whistling down upon us with hardly a moment's notice, except a hasty telegram from Washington with the familiar announcement that the thermometer will fall forty degrees in the next twenty-four hours. This is followed by nearly as rapid a rise in the temperature.

The extremes of our climate keep us house-bound for a large part of the year, and this is a very potent factor of ill-health. There are few portions of the country where it is agreeable to be out of doors for as many as a hundred days of the year.

The effect of a dry, variable, extreme climate is to stimulate powerfully the nervous system and keep it on the *qui vive*.

The liberty and enlarged scope of thought and action, which American women enjoy, must also be considered as causes of ill-health. How much more nervous energy an active, ambitious, American woman must expend than a German matron, with her placid, narrow life.

There are many other unavoidable causes of ill-health in American women which need not be discussed here; let us rather turn our attention to the more practical consideration of the avoidable causes of ill-health in our women. While there are very many of these, it has seemed to the writer that the following are the principal ones:

1. Lack of general exercise.
2. Lack of specific exercise.
3. Lack of abdominal breathing.
4. Improper modes of dress.
5. Superstition.

As every one is aware, there are three causes for the circulation of the blood.

(*a*) The contraction of the heart;

(*b*) Contraction of the voluntary muscles;

(*c*) The contraction of the diaphragm.

American women attempt to dispense with the last two; they neither take exercise nor do they breathe with the diaphragm. The muscles act precisely like the bulb of a Davidson syringe; when they contract the blood is forced into the veins; when they relax a new supply flows in.

Dubois-Reymond determined by experiment that the minimum amount of exercise, necessary to maintain the circulation, is equivalent to a walk of five miles a day. American women do not average one-fifth of this amount.

The diaphragm acts like the piston of a great pump, rising and falling sixteen times a minute, and pumping the blood out of the abdominal and pelvic cavities. Where its stroke is only one-half or one-third the normal, the amount of blood raised must be correspondingly less. The investigations of Dr. Thos. J. Mays, of Philadelphia, and of Dr. J. H. Kellogg, of Battle Creek, Mich., have shown that women ought to breathe precisely as men do. The thoracic type of respiration in women is entirely artificial, and is not, as physiologists have claimed, a wise provision of nature, having in view the restriction of the movements of the diaphragm during pregnancy.

The second cause assigned for the ill-health of American women is lack of specific exercise. All exercise is not of equal value; exercise of the arms and legs, while of great value, is relatively far less important than exercise of the trunk muscles, for the reason that the circulation through the thoracic, abdominal, and pelvic cavities is dependent upon exercise of the muscles surrounding these cavities. When the trunk muscles are not freely and systematically exercised, the circulation through the lungs and through the abdominal and pelvic organs becomes feeble, and the functions of these organs are imperfectly fulfilled.

We have now arrived at the point where we can trace, step by step, the evolution of ill-health in the American woman. Her undeveloped body is encased in corsets when she is fifteen years old. At school she learns a great many things, but is not taught that in order to have good health she must exercise the muscles of her body, and especially those of the trunk, daily and systematically. After marriage she settles down to a life of physical inactivity; she takes hardly any exercise, and even this little is not taken systematically; she does not breathe with the diaphragm; her circulation becomes feeble, her hands and feet are always cold, the blood accumulates in her abdominal and pelvic cavities, the functions of the abdominal and pelvic viscera are imperfectly carried on; she becomes dyspeptic; her stomach is distended with gas; her liver and intestines are torpid; the waste products of the system are not carried off, but accumulate in the blood. The opinion is constantly gaining ground that most of the functional nervous disturbances in women are due to auto-intoxication.

By-and-by the pelvic organs begin to show signs of disease. When one hears of the daily exploits of the abdominal surgeon, and learns that there is hardly one woman out of five who has not some form of pelvic disturbance, the conviction forces itself upon the mind that surely our women must be grossly violating some fundamental law of health. We have traced out the chain of physical causes which lead inevitably to a stasis in the abdominal and pelvic circulation. As a further result of this stasis there occurs a sagging of the abdominal and pelvic viscera, and as the latter are underneath, they catch the worst of it. Malpositions of the uterus are produced; the power of resistance of the pelvic tissues to invasion by pathogenic microbes is lowered; the tendency to plastic exudations is increased; the resolution of inflammatory processes is very much retarded; and thus the foundation for every variety of pelvic disease is laid.

The last cause assigned for ill-health in American women is superstition—not religious, but pharmaceutical superstition. The sublime faith with which an American woman will continue to swallow nauseous drugs in the belief that they will restore her to health and keep her in health, is only equalled by her faith in cosmetics. Drugs are wonderful things in their place, but it is not possible by any combination of drugs to replace the natural processes of health.

How are we to forestall this most serious of all our national evils, the physical degeneracy of American women? This matter is far too widespread and serious to be rectified by a little Anglomania, a little calisthenics, a little athletic craze. There are far too many women who lead lives that can only be characterized as parasitic; who have no independent existence, but merely cling to life like a polyp to a rock. They generate barely enough nervous energy to eat, drink, sleep, dress, take a great deal of medicine, complain constantly, and finally drop out of life without leaving the slightest vacuum.

The only way to meet this great evil is to introduce physical culture into our schools and make it compulsory. Not until then will our women realize the standard of feminine health as laid down by Walt Whitman:

They are not one jot less than I am,
They are tanned in the face by shining suns and blowing winds,
Their flesh has the old divine suppleness and strength,
They know how to swim, row, ride, wrestle, shoot, run, strike, retreat, advance, resist, defend themselves,
They are ultimate in their own right—they are calm, clear, well-possessed of themselves.

Louisville, Ky.

POSSIBILITIES IN THE THERAPY OF NUX VOMICA.

(*Second Paper.*)

By E. MACFARLAN, M. D.

Further clinical observations in the therapy of nux vomica have afforded me an accumulation of evidence of its efficacy in certain and hitherto untried affections of neurotic origin.

CASE I.—Was called, about one year ago, to Mrs. McG., age, 36. Found she had just miscarried at about four months pregnancy. The fetus and placenta had been expelled, but owing to inefficiency of uterine contraction, there was profuse hemorrhage; consequently soon after my arrival I found I had to contend with not only hemorrhage, but a grave case of syncope. Her extremities became cold, and the heart-beat was scarcely perceptible; and surely the case was rapidly becoming one of heart-failure. Quick action was necessary, and I at once concluded to test nux vomica, and proceeded immediately to administer one drop doses of the tincture every five minutes. When the sixth dose had been given the radial pulse-wave could be felt, and the heart-beat distinctly heard. After five more doses the pulse was nearly normal; warmth of body was restored and she was able to converse. Hayden's viburnum compound brought on good uterine contraction, and I succeeded in controlling the hemorrhage. In a few hours after the first dose of nux vomica I found my patient in such a comfortable condition of reaction that I left for home.

About six months afterward I was called in the early part of the evening to Mrs. McG., who believed she was again pregnant and threatened with miscarriage. She had pain in the back and profuse hemorrhage. I was getting good control of the hemorrhage when syncope set in, and on finding it was becoming protracted and her condition very similar to that for which I had used the nux vomica so successfully, I at once resorted to the same treatment and with the same satisfactory result.—But I must state, she was mistaken as to pregnancy. In the course of that night she passed a very large blood-clot which was the cause of her trouble; it evidently had been accumulating in the uterus through several menstrual periods.

CASE II.—In the fall of 1892, I was called to Rose L., (colored), age, 20. Her symptoms at first were of indigestion and sluggish liver, and congested portal circulation; but on further investigation of the case I discovered considerable tenderness over the left ovary. On the third day of my attendance, contrary to my advice to remain in bed and keep quiet, feeling quite comfortable in the afternoon, and getting permission of her mother, she left her bed to sit in an adjoining room. In the early evening of that day I was sent for in haste to see Rose. I found her in a prolonged state of syncope, and her mother was greatly alarmed, fearful she would die. The patient's pulse was small, weak and frequent, and counted 112; extremities cold; she was in a semi-conscious state, and when aroused would reply in a whispering voice, but immediately relapse into indifference to her surroundings. It seemed to approach so nearly to a case of heart-failure that I administered nux vomica in my usual way as to dosage and intervals, resulting within an hour in complete restoration of pulse, warmth of body and consciousness, and my patient became quite cheerful and talkative.

CASE III.—C. M., age, 28, had an attack, of acute inflammatory metastatic rheumatism. In the third week of the attack and while the disease was yielding to treatment, he was attacked with an exceedingly troublesome cough, but as he had been confined to his room, and in fact to the bed, by the rheumatism during this time, I was puzzled to account for the cough. Then I began to observe more closely the character of the cough and expectoration. A careful auscultation did not reveal any bronchial inflammation. By closely questioning the patient I found the cough was paroxysmal, and commenced with an annoying tickling in the throat; the cough steadily increased in violence with the return of every paroxysm until it became exhausting, and the patient would feel quite prostrated after each attack. I found also, his most severe paroxysms came on within ten minutes after his evening meal. I accordingly timed my evening visit for observation. I was now satisfied—having watched a paroxysm from beginning to end, which lasted over 30 minutes—that the cough was laryngeal and neurotic.

On examining the sputa it was found glairy and tenacious, and not such as we find in inflammatory conditions of mucous membrane of the air passages. I had given him a mixture which included compound tinct. benzoin and codeine without effect, but now being convinced the affection was of neurotic origin, it occurred to me to make a trial of tincture nux vomica in my one-drop-five-minutes-dosage. I timed my call the next evening so as to be present when the paroxysm of cough commenced. My patient had just finished his evening meal, and we had been conversing hardly ten minutes when the cough set in and rapidly increased in violence, and then came my opportunity for the crucial trial of nux vomica in this case. When I had given the seventh one drop dose the paroxysm began to yield, and by the time we had given the tenth dose it was almost over, the patient having an occasional cough. My patient was very happy over the sudden and unexpected relief afforded by the medicine I had just given him in such small doses, and of course was curious to know what it was. He then told me, these hard paroxysms sometimes lasted up to midnight, and on one occasion it had lasted almost until morning. I left him drops to take in the night in case he should have another paroxysm.

When I reached home that evening I do not think my patient was any happier than I was over my success with the nux vomica in a purely experimental case. The next morning Mr. M. informed me that he had slept through nearly the entire night, which, owing to the cough, he had not been able to do in a week. The patient from that night had no return of severe paroxysm of coughing. I furnished him with fresh drops every day, and he became so well posted in their dosage and use that in a few days the cure of the cough was complete. And I may here add that I have had several similar cases of cough since that, and the nux vomica has succeeded in all. If physicians would more closely diagnose their cases of cough, I will venture to say, there might be discovered more than one case of neurotic origin and laryngeal.

Case IV.—Mrs. L. S., age 40, is of nervous temperament and has occasionally attacks of a mild form of hysteria. About one month ago I was called to Mrs. S., who, I found, had the usual symptoms of “grippe.” I treated her accordingly, and she was convalescing nicely. I had made my morning call, and found her doing fairly well—excepting that she complained of a slight shortness of breath. I carefully examined her lungs, but there were no indications of pneumonia. In the afternoon, about 3 o’clock, I received a hurried message to come without delay to see Mrs. S. I found her in a very nervous, somewhat hysterical and desponding condition; her face was flushed, she had distressing dyspnea, cardiac palpitation, pulse 112, small, weak and frequent, which of course was due to the functional cardiac derangement—there being no indication of organic disease. She called my attention particularly to her difficult breathing so soon as I reached her bedside, and begged for relief. I took in the situation at a glance, and wasted no time in thinking of a remedy to meet the case. My previous experience, and knowledge of the efficacious therapeutic action of nux vomica in neurotic affections, prompted its immediate use in this case.

The fifth one drop-five-minute dose reduced the pulse to 100, and in a measure relieved the distressing dyspnea. When four more doses had been given the pulse dropped to 90, and the dyspnea was so nearly conquered that the patient expressed her thankfulness for such speedy relief after a steadily increasing suffering of over four hours. By the time four more doses had been taken respiration was normal and the pulse 84. My medical mission was accomplished, and I left my patient—who was now feeling very comfortable—with instructions to continue the one-drop doses every half hour for four more doses, and then every hour up to bedtime. Next morning she was sitting up in her bed feeling quite comfortable, with respiration normal and pulse 78, the palpitation having entirely subsided. She has had no further trouble up to date.

Case V.—About two weeks after the occurrence of this last, I was called to Mrs. E. C., age 31. I found her suffering from great dyspnea and palpitation. She was also greatly troubled with flatulent dyspepsia and frequent eructations. The difficulty of breathing and the palpitation of the heart had been increasing for several hours, and as it was a new experience in her life, she and her husband became greatly alarmed lest the attack should prove fatal. But I knew my patient, as she had been one of my clientele for many years. I knew she was of nervous temperament and slightly hysterical. I wasted no time, but administered tinct. nux vomica according to my usual plan; the particulars in this case would only be a repetition of others, and it is unnecessary to recite them. The result was an equally happy recovery.

314 W. 126th St., New York.

THE THERAPEUTICS OF STRANGULATED HERNIA.

By W. C. Abbott, M. D.

No doubt some will smile at the above caption, and say: “The idea! Who ever heard of such a thing?” Others will read what follows and, perhaps, profit thereby.

Strangulated hernia is looked upon as, strictly, a surgical affair, only two ways of treatment being thought of, *i. e.*, reduction by taxis or, in case of failure—and if the patient and family will consent, the knife. The former usually fails, and consent to the latter is frequently so long deferred that the result is anything but satisfactory; hence, any measure which is at the same time simple and helpful should be looked upon as a valuable acquisition.

The plan of treatment which I wish to present will best be understood if illustrated by the record of a case in hand. One morning, recently, I was summoned hurriedly to see an elderly German laborer, whose “breach was down,” so the messenger said, and it surely was. Such a scrotal display I never saw before; a mass larger than a child’s head, and as hard and tense as a foot-ball. I thought, at first, that part of it must be an old hydrocele, but was assured by the sufferer that “him all goes back.” This rupture was of some ten years standing. Formerly the hernia was retained with a truss, but of late years it had been allowed to go up and down according to circumstances, never having become strangulated before. Now, my experience has been that when an old man who has had years of practice reducing his hernia fails to put it back, the surgeon has a job on his hands.

In this particular case most of the mass was resonant, but there was a lump the size of a goose-egg, close to the ring, that was hard and gave a perfectly flat precussion note—the omentum—congested by strangulation, the very irritation from which caused a tighter spasm of the muscle-fibres between which the mass protruded. I tried taxis faithfully; so had the old man, and we both failed. I used hot fomentations to no avail; I placed him at an incline of 45° and sprayed the mass with cold water to stimulate contraction of the muscles of the cord. They contracted sharply, but had no effect except to increase his pain. I lowered him to the horizontal position, covered him up with a hot blanket, and sat down to rest.

Here was an old man, hernia strangulated for some two or three hours, a serious case. It was growing worse every moment, and nothing seemed likely to avail but the knife. The spasm was intense. How could I relax it?

I hesitated to try an anesthetic for various reasons, one of which was that *I found a piece of candle burning in a beer mug at the feet of my patient*. Knowing the great value of hyosciamine in spasm of the viscera, I took from my ever-present alkaloidal granule-case hyosciamine amorphous, gr. 1–134, and gave it hypodermatically with morphine sulphate, gr. ¼, and atropine, gr. 1–250. In a few moments my patient said, “that’s better”; and in less than ten minutes a gurgling of gas was heard through the mass indicating relaxation of spasm. Thus encouraged I gave a second dose, adding to hyosciamine, gr. 1–250, strychnine arseniate, gr. 1–48, the latter to induce forced peristalsis. In less than ten minutes more, I heard a loud gurgling sound, and my patient cried, “him’s gone,” and, sure enough, it had. The entire mass had disappeared through a hole the size of a nickel.

A retentive bandage for the time, and a well fitted truss a few days after, completed the treatment of, to me, an interesting case, which leads me to suggest a “therapeutics for strangulated hernia” for your consideration.

2666 Commercial St., Chicago.

PERISCOPE OF THERAPEUTICS.

By J. Lindsay Porteous, M.D., F.R.C.S., Ed.

Methylene Blue.

Recently I have had an opportunity of proving the efficacy of this new solvent, and am much pleased with the result obtained.

During the months of August and September I had five children under my care, suffering from diphtheria. As all the children belonged to one Institution, had the same nurse, and in every way the same surroundings, I had a good opportunity of testing the solvent powers of this drug, as compared with others. In the first child, age 4 years, I used acetic acid spray as a solvent, giving iron and potassium chlorate internally. Although the membranes in fauces visibly shrank, the disease extended downwards rapidly, almost choking the patient. Intubation was resorted to, which for a time relieved the choking symptoms, but the child gradually, sank.

Two more children were seized with the disease and the fauces were painted with sulpho-calcine, a remedy I have used successfully for some years, and corrosive sublimate was given internally. The children improved, but the patches were a long time in shrinking. I applied an aqueous solution of methylene blue (1:9) to the patches by means of a camel’s-hair brush, three times a day. The effect was remarkable. Not only did the patches, after the third application, shrink, but the feverishness and restlessness abated. Three days after the commencement of the treatment all the patches in both children had disappeared.

Two more children, aged 5 years and 3½ years, were attacked. As the patches covered the uvula, tonsils and back of the pharynx, and no further down, I thought that I would try local treatment only. After a few days the patches disappeared, the fever left, and the children were convalescent.

Piperazin and Piperazin Hydrochlorate.

I have of late had ample opportunity to test both piperazin and piperazin hydrochlorate as solvents of uric acid. Hitherto I have used the salts of lithium and potassium in the uric acid diathesis, but will now certainly always use the new solvent in preference to either of these. I have taken particular pains to watch the action of the drug—so feel confident in stating that it is the best of known uric acid solvents.

I have never noticed any toxic effects from the use of either preparation, but prefer the hydrochlorate on account of being less hygroscopic, although I believe piperazin can now be procured in lozenge-form in glass tubes, which greatly lessens the hygroscopicity.

I will now briefly give particulars of three cases of uric-acid diathesis treated with the medicine.

The first, a man aged 36, weight, nearly three hundred pounds, suffered much renal pain, with occasional symptoms of small calculi passing along ureter. The urine had very acid reaction. The microscope revealed numerous crystals. The murexide test likewise proved that uric acid was present in abundance. This patient, wishing to become thin, was put on a beef diet, no potatoes, few vegetables, etc., just the diet to increase the uric acid. He was put on the solvent treatment, giving twelve grains, divided into three doses daily. Each dose was given in a tumblerful of vichy water. In a few days the urine became neutral in reaction; the loin-pains left; also the constant desire to micturate caused by the irritation of the uric acid. The beef-treatment was kept up for several weeks and the weight was reduced by 11 pounds. This case shows that, although no precautions were taken to diminish the formation of uric acid, but the contrary, piperazin was sufficient to dissolve the crystals.

The second case was that of a man aged 50, weight, 215 pounds. He complained of pain, described as “burning” in loins; also a dull ache in bladder, frequent desire to micturate; sometimes urine was mixed with blood. Without altering the diet at all, which was always a liberal one, including beer every day, all the disagreeable symptoms disappeared while taking piperazin hydrochlorate. On several occasions there was a return, in a modified form, of the symptoms, but after a few doses of the drug they always disappeared.

The third case was that of a man 60 years of age, who had for years felt pain in kidneys, along ureter and in bladder, and never got much relief. I prescribed for him 5 grains of piperazin to be taken three times daily in large quantities of water. After a week’s treatment the pains had left, and he was better than he had been for years.

Gruber has used it with good results in diabetes. He considers that the action of the drug in this disease is inhibitory to the transformation of glycogen into sugar.

Hypodermatic Alimentation.

Caird (*Edin. Med. Journal*, September, 1893) reports a case of extreme weakness and emaciation due to malignant stricture of the esophagus, which was improved by intra-muscular injections of sterilized olive oil. In the course of a week from three to four ounces of oil were injected into the gluteal region. There was no pain or inconvenience caused by the injection. Sugar was occasionally combined with the oil. None of the skin punctures inflamed. There seemed no limit to the amount of oil which a patient can tolerate.

Yonkers, N. Y.

Cold Due to Bacteria.—Bacteria are likely to be blamed for all the ills that flesh is heir to. Prof. Schenck now maintains that what we call a “cold” is really due to these invisible pests. When one enters a cold room after being heated the bacteria in it flock to the warm body and enter by the open pores of the skin. Whatever may be said of this hypothesis he seems to have proved by experiment that bacteria in the neighborhood of a warm body move toward it. The confirmed smoker may derive some comfort from the fact that tobacco is inimical to them.—*London Globe.*

Recent Medicaments.

A Decade of New Remedies.—"There are at least one hundred new remedies of synthetic origin now in general use that were not known ten years ago." So an American contemporary says; but we doubt if anyone can compile a list of one hundred. Who will try?—*Chemist and Druggist* (London), Sept. 30, 1893. We are not responsible for the original statement; but the invitation is general, and we are anxious that our English friend should have a prompt response from America, so here goes: Acetanilid, agathin, alpha-oxynaphthoic acid, alumnol, amyl-enhydrate, analgen, antipyrine, antiseptol, antispasmin, anti-thermin, aristol, asaprol, asepsin, benzanilide, benzonaphthol, benzosol, betol, bromal-hydrate, bromoform, bromol, chinoline, chloralamide, chloral-ammonium, chloralose, chlorphenol, creolin, cresalol, cresin, cresol, cresol-iodide, cresotic acid, diuretin, dulcin, ethyl bromide, ethyl chloride, eugenol, eugenol-acetamid, euphorin, europhen, exalgine, formalin, formanilid, gallacetophenone, gallobromol, gallanol, guaiacol-carbonate, homatropine, hydracetine, hydroquinone, hypnal, hypnone, ichthiol, iodol, iodopyrine, kairin, losophan, lysol, metaldehyde, methacetine, methylal, methyl chloride, methylene-blue, methylene chloride, methyl-violet, microcidine, naphthalene, naphthol, naphtopyrin, orexine, oxychinaseptol, paraldehyde, pental, phenacetine, phenetol, phenocoll, piperazine, pyridine, resorcin, resorcinol, saccharin, salacetol, salicylamide, saliphen, salipyrin, salocoll, salol, salophen, saprol, solutol, solveol, sozal, sozoiodol, styracol, sulphaldehyde, sulphaminol, sulphonal, tetronal, thalline, thermifugin, thilanin, thioform, thiol, thiophen, thioresorcin, thiosinamin, thymacetin, tolypyrin, tolysal, tribromphenol, trional, tumenol, uralium, urethane. This makes 114 new definite chemical products—and the list is not exhausted. It may also be urged that some of these products have been in use longer than ten years. This is the case only with naphthol, which was first recommended as an antiseptic in 1881; but we have omitted several legitimate new naphthol compounds, so that a balance is maintained. Other products, as acetanilid, guaiacol, anilin colors, pental, etc., were known as chemical products; but their medicinal application dates back less than ten years.

Referring to above list, a casual examination shows, that about thirty-three are patented products; between fifty-five and sixty bear proprietary "utility" names (including the thirty-three patented); and about thirty are absolutely free and non-proprietary. We may publish an accurate division with details on some future occasion.—*Notes on New Remedies.*

LLARETA; A NEW ANTIGONORRHŒIC.—Dr. Infante (*Aerztl. Rundschau*). Llareta is the abbreviated name for *Haplopapus Llareta*, a plant growing abundantly in Chili, and with which the author claims to have obtained a radical cure within ten to fifteen days in every case of gonorrhœa in which it was tried. The following is his formula:

Fluid Extract Llareta	1 part.
Distilled water	30 parts.
Tablespoonful twice daily.	

—*Amer. Med. Surg. Bulletin.*

DISINFECTIN.—(*Pharm. Zeitschr. f. Russl.*) Disinfectin is the name of a preparation intended for ordinary disinfection, said to be obtained as follows: 5 parts of the residue left in distilling crude naphtha are thoroughly mixed with one part (by volume) of concentrated sulphuric acid, and allowed to cool. The fluid portion is separated from the sediment, and gradually mixed with an equal volume of ten per cent. soda solution, and well shaken. Thus is obtained a yellowish-brown emulsion,—disinfectin,—which, when intended for use, is diluted with four parts of hot water, and thoroughly shaken.—*Amer. Med. Surg. Bulletin.*

HIPPURIC ACID AS A DIURETIC.—This acid, obtained[1] from the urine of the cow, is a favorite diuretic with many French practitioners. Dujardin-Beaumetz prescribes it combined with lime:

℞	Hippuric acid	25 grammes

Milk of lime sufficient to neutralize it.	
Simple syrup	500 grammes
Essence of lemon to flavor.	

Four to six tablespoonfuls daily. As before mentioned it is excreted in the urine as benzoic acid.—*Provincial Medical Journal.*

THYMOL IN TOOTHACHE.—Dr. Hartmann (*Deutsche Med. Wochenschrift*) has employed thymol in toothache from hollow teeth, in place of arsenious acid. He fills the cavity of the tooth with a tuft of cotton on which a few crumbs of thymol have been sprinkled. It does not irritate the mucous membrane of the mouth much, and it is easily removed by rinsing the mouth with water. If a rapid action is desired let the patient rinse the mouth often, with warm water, in order to facilitate the solution of the drug. It never increases the pain at first, as arsenic does, and is not poisonous.—*Lancet-Clinic.*

CHLOROFORM NARCOSIS.—Resultation in chloroform narcosis has been accomplished by a new method devised by Maas, one of Koenig's assistants of Goettingen. In the first case, the ordinary means of resuscitation had been tried for an hour without effect: respiration and pulse has entered ceasec. Maas then made rapid rhythmical compressions, about one hundred and twenty per minute, of the cardiac region, whereupon the heart's action gradually increased and the patient recovered. In a second severe case responded with the same result to the treatment. Maas ascribes the effect of the cardiac compressions to the driving of the blood into the larger arteries. —*Chicago Medical Recorder.*

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Editorial.

INFLUENZA AND ITS TREATMENT.

The manifestations of Influenza in this section of the country during the past two months have been numerous and varied, although the malady has been far less directly fatal than on its first appearance in the winter of 1889–90. The different types of the disease, the pulmonary, the nervous and the abdominal, have been less distinctly marked, but it has been especially severe in the case of elderly people when appearing as an intercurrent affection. Another noticeable feature is, that the disease has shown a greater disposition to attack children than in former years, although when children have been enjoying usually good health, fatal results have been rare. This, however, has not been the case with sickly or puny children, as the disease has manifested itself in various ways, such as throat, ear and other complications. In many cases nothing more than the peculiar pains characteristic of the disease have been noticed; in others, it has passed off with nothing more serious than would result from a bad cold, so that large numbers have fought it out on this line. The prevalence of the malady in the vicinity of Philadelphia, and we presume this observation holds good elsewhere, has been wide spread, showing conclusively that it is largely dependent upon atmospheric influences. Indeed, up to the present writing, the condition of the weather has been extremely favorable to the development of influenza, and we all hope that better weather during the remainder of the winter will tend to check its spread.

Judging from published reports, the treatment of influenza during the present epidemic has been mainly symptomatic, simple remedies being employed in place of the powerful antipyretics and analgesics that were so extensively used three years ago, and this may account in part for the reduced mortality. In this connection the writer ventures to suggest the use of a limited number of remedies that have proven of signal service within the past two months. In the pulmonary type, to relieve the distressing and frequent cough, nothing has given better results than morphine hydrochlorate and pilocarpine hydrochlorate, given hourly in doses of one-fiftieth of a grain each, together. This combination seems to allay sufficiently extraneous irritation, while it favors the re-establishment of the normal secretions, and has shown remarkably favorable effects when broncho-pneumonia threatened to supervene. Given in these small doses, it produces neither narcotism nor depression, and the patient is ready to take his regular meals, to which is added liquid nourishment during the intervals. Hot milk—not boiled—and the free use of beef-tea made from a good extract of beef, are helpful in assisting to maintain the strength. In the abdominal type, with much pain along with mucous discharges from the bowels, showing involvement of the liver, a combination of mercury biniodide with codeine sulphate appears to control and modify the progress of the disease. One one-hundredth grain of the former with one-tenth to one-fifth grain of the latter may be administered every two hours, and along with this, where we have to contend with the pains peculiar to influenza—rheumatic and neuralgic—it is well to administer conjointly small doses of bryonia. From two to five drops of the tincture can be given at intervals of two to four hours. Cases which do not respond readily to bryonia, will often quickly show amenability to the administration of rhus toxicodendron, given in one-drop doses of the green root tincture at intervals of two hours. It is remarkable what power these two simple remedies exercise over the fugitive neuralgic pains peculiar to influenza, doubtless because they modify the nutrition of the cells composing the fibrous structures, enabling them to throw off waste products and thus maintain a condition approaching that of health. The after-treatment will embrace the administration of the arseniates of iron and strychnine, and in debilitated subjects this should be supplemented by the exhibition of cod-liver oil or petroleum in the form of emulsion.

It will be noted that nothing has been said in regard to the advisability of antiseptics, and for this reason, *viz.*: That although the disease may apparently be associated with a micro-organism, this microbe plays no important part in the various manifestations of the disease. Whatever influence it may possess is, as we have seen from clinical experience, counteracted, discounted, by what the older physicians were pleased to term the *vis medicatrix naturæ*; and besides, we have absolutely nothing to warrant us in assuming that our present antiseptic measures and remedies exercise any perceptible influence when taken into the system, at least so far as regards this particular malady. Further investigations in this line may develop some new ideas in this respect, but for the present, we must rest content with the stern facts as we see them at the bedside. Special attention should be directed here to the theory of "*Digestive Leucocytosis*," as elaborated by Professor Chittenden, published in another department of this number.

AUTO-INFECTION IN ABDOMINAL DISORDERS.

In view of the complications arising in the course of abdominal diseases from auto-infection, and with our recent knowledge in regard to this important but insidious factor, it behooves the physician to be on the look-out for such manifestations. Toxic anemia, as portrayed by Dr. JOHN E. BACON, in our last number, brings to light some valuable truths which should ever be uppermost in the minds of those having to deal with occult affections.

Considering that we now have ample evidence that a sick person may be his own worst enemy, by reason of an unhealthy condition of the alimentary canal, it is not too much to assume that in many cases of prolonged illness, the complications arising may frequently be due to auto-infection, and thus, in addition to the disease which is seen, there is another disease engrafted upon the first which is insidious but persistent. The plan of using purgatives to unload the intestinal tract and relieve the portal circulation has its value, but the habit, once established, cannot be relieved permanently by this method. The addition of antiseptics is likewise a valuable feature in the treatment of this class of cases, and the two combined will often serve a useful purpose; but cases occur in which either method alone, or both combined fail, and it is then that we are compelled to study the philosophy of cell-function.

In this connection, the studies of POHL, as elaborated by CHITTENDEN, promise to shed a flood of light upon the vexed and complicated problem which involves the theory of nutrition. If digestive leucocytosis, as delineated by these authors, be true, and we have no reason to doubt their conclusions, then we have a satisfactory explanation of the value of proper food for the sick, to say nothing of the nutrition of those who ordinarily enjoy good health. The fact being admitted that white blood-corpuscles or leucocytes are more rapidly developed after the ingestion of food, it follows that this cellular activity was intended to accomplish or aid in accomplishing certain metabolic changes. And when to this fact we add the knowledge advanced by METSCHNIKOFF concerning the function of phagocytes, which are said to be modified white blood-corpuscles, together with the scientific demonstrations of VAUGHAN, that these cells produce through the activity of their nucleus an antiseptic substance, we have a complete scientific explanation of the need for suitable food stuffs, not only in illness of every description, but also as a precautionary measure against disease.

It is time that the study of the class of cases under consideration should be placed on a scientific basis, in order that we may have some definite idea of the objects to be attained—and the method of attaining them—expressed in terms which may be comprehended by the merest tyro in therapeutics, and to this end the contributions of METSCHNIKOFF, POHL, CHITTENDEN and VAUGHAN must be accepted as valuable preliminary data. The next thing in order will be to make these scientific facts clinical facts by making them practical, and it is therefore necessary that others should take up the task in order that the work may be carried to a successful termination.

THE PRINCIPLE INVOLVED IN THE SUBCUTANEOUS USE OF BLOOD-SERUM.

An important revelation has been made to the medical profession in the form of a communication to the *Medical News* (Jan. 13, 1894), by Dr. C. F. DARNALL, of Llano, Texas. According to the report, twins suffered from ptomain-poisoning, and one of them died. The surviving child was treated by the subcutaneous employment of a normal salt solution, but without apparent benefit. Later, this was followed by the use of two ounces of blood-serum drawn from the arm of the father, a healthy young man. To quote the words of the author, “The child was carefully watched, and reaction occurred in about six hours. Improvement gradually took place, and in six days, upon a diet of thin corn-meal gruel at first, later by fresh milk prepared, the child was well. It was placed at the mother’s breast at regular intervals, and at the time of writing, nearly nine weeks after the onset of the illness of the children, the lacteal secretion is fully established.”

Now, this is evidently a very interesting case, and all physicians who wish to understand the “whys” and “wherefores” would like to have some additional light thrown upon this occult subject. In this instance, there is no intimation that the father had been previously rendered “immune” to the peculiar ptomain which, in this case, is supposed to have been derived from a can of condensed milk. How, then, could this blood-serum have exerted a favorable change in the metabolism of the infant which was but ten weeks old? Our bacteriological friends will tell us that the benefits were due to the “*natural antiseptic properties*” of normal blood-serum; but the clinician and intelligent and conscientious physician will want to know from whence this peculiar property is derived. The blood is an alkaline fluid, and we are taught that antiseptic solutions, to be effective, must be acid. How does it happen, therefore, that blood-serum obtained from normal blood, which is alkaline, possesses antiseptic properties?

This question has already taken up so much space in the journal that, aside from our prospective subscribers, it would not be advisable to take further time for its elucidation. It will be sufficient, however, to say that it involves a principle first developed from a scientific standpoint by Professor VAUGHAN, who has shown conclusively that the antiseptic property of blood-serum is due to the fact that the nucleus of the white corpuscles secretes an actively antiseptic substance, a substance having all the characteristic of a proteid, which he has denominated “*nuclein*.” At the risk of becoming monotonous in furthering this measure, we repeat that nuclein solutions may be obtained from yeast-cells, from the yelk of the egg, from the thyroid gland, from the spleen and from other organs of the body, and when properly prepared, they are as powerful in modifying the multiplication of microorganisms as is corrosive sublimate or any other recognized bactericide. The principle underlying the subcutaneous use of blood-serum is, therefore, demonstrable, and is strictly within the confines of scientific medicine. How long it will require to educate the medical profession in this knowledge remains to be seen; but if it was merely an empirical claim or chimerical fancy, and backed by sufficient capital or government patronage, the period would be short indeed.

EDITORIAL NOTES.

The contents of this issue of the AMERICAN THERAPIST are as usual varied, practical, interesting, original—carefully arranged to make the reading harmonious as well as profitable. We say this for the benefit of those casual readers of this issue who are not yet—but ought to be—subscribers and regular readers.

The American delegates to the International Sanitary Congress, to meet in Paris this month, were named by the chief officer of the Marine Hospital Service; the delegates are Dr. Stephen Smith, New York City, Dr. Shakespeare, Philadelphia, and Dr. Bailhache, Washington, D. C.

The advance copies of the "Minutes, Reports, Papers and Discussions of the 41st annual meeting of the American Pharmaceutical Association, held at Chicago, August 14th to 20th, 1893," have just been issued to members of the association. The regular bound volumes of the "1893 Proceedings," containing the notable annual report on "Progress in Pharmacy," will be issued later.

THE NEWBERRY LIBRARY.—Prof. Senn has made a munificent gift to this institution by giving to it his collection of medical books, including especially valuable works on anatomy and surgery, full sets of periodicals, and the collection of books of the late Professor Baum of Göttingen. With this nucleus, added to by other donors, and its own already extensive collection and ample resources, the Newberry Library is apparently in the front rank, and will afford the profession in Chicago unexcelled bibliographical facilities.

The four years' course is gradually being adopted by all the leading medical colleges. The Jefferson Medical College (Philadelphia) has just concluded to make the four years' course obligatory after this year, and the Medico-Chirurgical College (Philadelphia) is considering the advisability of making the same rule. The latter college, by the way, has just established new professorships of Otology, Genito-Urinary Diseases and Orthopedic Surgery; the new offices will be filled shortly.

The PENNSYLVANIA STATE MEDICAL SOCIETY will hold its next meeting at Gettysburg, May 15, 16, 17, and 18, 1894. Those desirous of presenting papers are requested to notify, at an early date, the Chairman, or any other member of the Committee of Arrangements. Dr. E. E. Montgomery, of Philadelphia, is the chairman, and the other members of the committee are: Dr. Isaac C. Gable, of York; Dr. Geo. S. Hull, of Chambersburg; Dr. John C. Davis, of Carlisle; Dr. Henry Stewart, of Gettysburg; Dr. George Rice, of McSherrystown; Dr. E. W. Cashman, of York Springs.

The American Medical Association meeting is to be held this year at San Francisco, on Tuesday June 5th. To Californians this date is a little late for showing off their City and State to best advantage; it will be a trifle too hot for comfort by that time. A better date would have been May 5th, and by arranging early excursions the visitors could have taken in Lower California in April, then the Mid-Winter Fair—which will undoubtedly extend to May—and wound up their hours with attendance at the Convention. Perhaps a general request to re-consider the date should result favorably with the Committee who make the date and all arrangements.

After a spirited and quite acrimonious campaign, the New York County Medical Society held its annual election Monday evening, January 15th, 1894. All the candidates on the regular ticket, excepting Dr. McLeod for president, were elected on the first ballot. For the presidency three candidates were in the field, and none of them obtained a clear majority on the first ballot; a second ballot resulted in the re-election of Dr. McLeod. The candidacy of Dr. Alexander, urged by the younger and progressive members, was the disturbing cause in this election; the feelings of the contesting factions were wrought to a high pitch, and the proceedings were so turbulent that the daily press of New York took up the matter and regaled the public with entertaining accounts of the contest. All is now harmony again, we hope.

Correspondence.

THE DOSE OF SANTONIN.

To the Editor:

Sir:—On page 678 of the Cincinnati *Lancet-Clinic*, of Dec. 9, 1893, appears a letter from J. M. Murkon, M.D., of New York, to the editor, calling attention to the dose of santonin, used as an emmenagogue, which I had reported in the American Therapist, vol. I., no. 1, p. 8 (July, 1892). Dr. Murkon asks "whether the dose as given is not a misprint; if not, it is certainly more than dangerous, as five grains has caused a fatal result."

I wish to say that in the year and a half since the article in question was written, I have often made use of *ten-grain* doses of santonin for its emmenagogue effects, and in no instance have I learned of the slightest discomfort from such a dose. I rarely administer more than a single dose; but if more are required, I invariably wait twenty-four hours before giving them. The dose is always administered at night on retiring, and at the same time I order a mustard foot-bath, and frequently hot drinks *ad libitum*. I would say again that I have never seen this procedure lead to any untoward effects, or even any discomfort, and usually the menstruation is comfortably and apparently normally established by the next day.

I do not believe it is possible to produce miscarriage or abortion by this treatment. I have never seen it follow, and so certain am I of this fact that I have come to look upon it as being a safe and convenient means in making a differential diagnosis as to the presence or absence of conception in cases of suppressed menses. I am so confident of this diagnostic value that I am in the habit of saying to the patient when there is any doubt as to the cause of the suppression, "I shall give you a dose of medicine which will bring matters around all right if you have taken a cold, but if there is a natural cause for the arrest of the menstruation you need look for no results." Who has not occasion, frequently, to decide for the anxious patient the *cause* of delayed menstruation? They come to you insisting that the suppression is not due to conception, and are exceedingly importunate in their demands for relief. Here, then, is a medium through which I have frequently solved this perplexing question; and through which I have been enabled to conscientiously *prescribe* for such cases, while otherwise I should have refused to use any means to afford relief, simply advising patience, and awaiting the tedious restoration of function by Nature's own forces.

D. H. Bergey, M. D.

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ICE IN BRONCHIAL ASTHMA.

To the Editor:

Sir:—In the November number of the American Therapist, I contributed a short account of a long, violent and very refractory exacerbation of bronchial asthma, which yielded almost instantly to the application of ice-packs over the pneumogastrics. The vagi are held to contain both dilator and contractor filaments for the bronchial muscles. This attack seemed to me to be a convulsion, so to speak, of the afferent contractor filaments. By freezing the vagi, then, these violent motor impulses should be inhibited; and such turned out to be the case.

A day or two since I received a letter from Dr. Ezra Peters, of Missouri, saying: “I have just returned from a case of asthma very similar in every particular to the one reported by you in the November number of the American Therapist, except that I did not use nitro glycerin. After a hard tussle for fifty-two hours for air, a fifteen minute’s application of the ice-pack to the neck caused the respiration to become quite full and free, and the pulse to fall from 138 to 80.”

I publish this additional experience in the hope that others will give this simple procedure a further trial.

Ernest B. Sangree, M. D.

2020 Arch St., Philadelphia.

Current Literature.

NATURE'S CURE OF PHTHISIS.

Dr. Henry P. Loomis states (*Med. Rec.*) that he has found quite a number of cases of recovery from phthisis. His summary is as follows:

1. Out of 763 persons dying of a non-tubercular disease seventy-one, or over nine per cent., at some time in their life had phthisis, from which they had recovered.

2. The new fibrous tissue by which the advance of the disease was apparently checked and the cure effected, developed principally by round-cell infiltration of the interlobular connective tissue, which in some instances had increased to an enormous extent. Some of the new fibrous tissue was formed later by round-cell infiltration in the alveolar walls and around the blood-vessels and bronchi. Pleuritic fibrosis appears to be secondary to tubercular processes in the lung substance. The interlobular connective tissue is the primary and principal source of the fibrosis.

3. Tubercle bacilli were present in the healed areas in three out of twelve of the lungs examined. These healed areas did not differ in their gross or microscopical appearances from those in which they were not found.

4. Thirty-six per cent. of all cases where the lungs were free from disease showed localized or general adhesions of the two surfaces of the pleura.

VENTRO-FIXATION OF THE UTERUS.

Dr. Spaeth, of Hamburg, according to the *Lancet*, has now published reports of twenty-five cases in which he has performed the operation known as ‘ventro-fixation of the uterus.’ None of the cases proved fatal. In seventeen permanent anteflexion was obtained; in fourteen there was, besides the retroflexion, a diseased condition of the uterine appendages necessitating their removal. Of the cases that were not so complicated all except one were successful. Dr. Spaeth rarely fastens the stump of the broad ligament into the abdominal wound, usually stitching the fundus uteri directly to the parietal peritoneum. In the later cases he adopted Schede’s method—that is to say, silver sutures were drawn through the whole thickness of the abdominal walls at intervals of about an inch and a half, but they were not at first tied. In the intervals finer silver sutures were inserted through the sheaths of the recti, the peritoneum and the fundus uteri, and tightened, twisted, and cut short, the whole of course being beneath the skin; the thicker sutures were then tightened and twisted and the lips of the wound brought together with superficial catgut sutures. The subcutaneous silver sutures remained, but never gave any trouble. Dr. Schede and Dr. Spaeth are both of opinion that this method is the best for preventing any hernia, and that when it has been employed abdominal binders are unnecessary. Dr. Spaeth does not perform or recommend ventro-fixation in cases of retroflexion unless there is either disease of the appendages or chronic peritonitis.

PNEUMONIA.

Croupous pneumonia has during recent times been defined as “an infectious disease characterized by inflammation of the lungs and constitutional disturbance of varying intensity.”

The recognition of an infection as a cause of this disease necessarily implies the existence of a specific germ, and as corollary we may state that this germ, like all germs, has a limited existence.

During the cause of the inflammation three stages have been recognized—congestion, red hepatization, and gray hepatization. Now, on a physician being called to a case of this kind, the query is, what is the best thing to do in the way of medical treatment?

If one consults some of his associates he will find remedies recommended *ad nauseam*. One declares the fever does no harm, while another is not satisfied unless one of the chemical antipyretics keeps the temperature near the normal standard.

Varying results, of course, follow these different treatments.

Having for a long time accepted and practised the doctrine that many diseases can be modified and conducted to a safe issue by giving a remedy that will attack the diseased cell-function, I have treated my cases of pneumonia in this way. The three remedies that I employ are aconite, bryonia and iodine. If I am called to see a patient during the first stage I give aconite and iodide of potassium, and on the development of the second stage I continue the iodide and also give bryonia. The aconite is given with a view of relieving the vaso-motor tension, and thereby equalizing the circulation; this, if doing nothing more, produces a measure of comfort. The last two remedies are given for their local effect. Now, as to a dose: in a half goblet of water I put ten drops of tincture of aconite, and in another goblet containing same quantity of water I put five grains of iodide of potassium, and give a teaspoonful alternately every hour. As soon as the second stage comes on I substitute five drops of tincture of bryonia for the aconite and continue to give in alternation with the iodide. If the heart becomes weak give one-fourth of a grain of extract of nux vomica every three or four hours. If the patient has much pain, hot applications are used. This with suitable food constitutes a treatment that bears excellent results. The physician who believes in giving large doses of quinine, ammonia, etc., will sneer at these small doses and hotly declare that nature cures independently of the drugs. In view of this declaration there is a compliment paid to *vis medicatrix naturæ* that the writer recognizes, and to have such an ally is certainly a desirable help. Those who have faith in heroic dosage are not always mindful of this fact, nor yet of the condition which obtains under their hands, and that is a state of drug disturbance plus the disease. Nature here is hindered, not assisted.

S. B. CHILDS, M. D.,

The Brooklyn Medical Journal, Jan., 1894.

DRUG ACTION.

Chemistry makes such rapid strides of late that it is impossible for the medical man in ordinary practice to keep pace with it. We have ptomaines and leucomaines, and in the literature of the journals these two terms are used indiscriminately, which is unfortunate, as tending to create confusion. While ptomaines are those alkaloidal products of metabolism belonging to the cadaver, or to any dead organism, whether animal or vegetable, the term is now applied to the same products in living tissues and which really are not ptomaines, but leucomaines. Chemically speaking, both ptomaines and leucomaines exist in the forms of monamines, diamines, and triamines, and have hitherto been regarded as resulting from the oxidation of ammonium salts by abstraction of water or from acids by substitution of amidogen. Yet the bacteriologists assure us that they are the secretions, or rather excretions, of bacteria, just in the same way that the strata of non-igneous rocks of the earth are formed primarily by the secretion, and secondarily by the excretion, of calcareous shells of diatoms, or of the cretaceous shells of those protozoans entitled rhizapods. Elsewhere in the SUMMARY I have expressed myself to this effect in nearly the same language, and while it may seem pedantic, it should be borne in mind that this is the age of terminology.

We, each of us, have our special ties—shall I say hobbies? One of us making the eye a specially will make use of ophthalmological terms. Another turns his attention to the technology of animal tissues, and still another, with an eye towards the bacteriological existences, leads us into a labyrinth of Greek and Latin technics.

We are talking of ptomaines and leucomaines, and that their existence is probably due to bacteria. These are regarded as toxines, and certain diseases are said to be caused by these toxines, and not from the bacteria themselves directly, but indirectly. I appreciate the idea that the curative results of these diseases are due to the antagonism of the remedies to these toxines, and not to the destruction of the bacteria. Before we inquire into the curative action of remedies we must first find out how these toxines produce disease. The tendency is to trace the origin of all disease to a change in the protoplasmic cells. This change is evidently due to a toxine. A healthy physiological change in the cells may be termed metabolism, while an unhealthy or pathological change may be termed katabolism. The question arises, in what way does katabolism arise from these toxines? It occurs to me that it must be due to chemical affinity, or as it is sometimes very properly called, *chemism*.

In a brief but very excellent article on therapeutic action by Dr. Thomas J. Mays, of Philadelphia, in the AMERICAN THERAPIST, he commences thus: “Every phenomenon in nature becomes intelligible only when considered in the light of force. Any scientific system of therapeutics must therefore be built on a broader basis than that of the mere drug action on the animal economy.” In this he is unquestionably correct; but when he, along with others, asserts that drug action is due to what is termed “interference,” to me, at least, he becomes unintelligible. I am utterly unable to take in the idea. When he refers it to molecular motion, which, after all, is chemical change, he is right. Heat, light, electricity, motion—each interchangeable, correlating with each other, constituting force—must be the source of all chemical action. These are, strictly speaking, proteodynamics. They act, as I have indicated, upon the cell. This action may be either metabolic or katabolic, and when the remedial agent is presented the change results. So true is this that the *vis vitæ* itself results from chemism, so that every action, every change, has simply two factors—matter and dynamism.

In this connection permit me to make the following quotation from the *Medical News*: “Of course, it had been recognized that certain diseases are self-limited, and the phenomena of natural and acquired immunity were duly appreciated; but it required the knowledge gained by the advances in bacteriology to afford a rational explanation of these phenomena. There is yet much to learn. The beginning has but been made. Enough, however, has been seen to teach that disease has its chemistry, and that the treatment of the future will depend upon a knowledge of this fact and the application of chemic laws."

The progressive physician does not believe that medicines *cure*. It is true he uses the word just as we say the sun rises and sets when we know that it does neither the one or the other, but that the phenomena are due to the revolution of the earth on its axis. The cell function is the building function, you may call it metabolism or anabolism. It is either a building or a repairing process. Now derange the process by means of toxines, and you have what may be called katabolism. The molecules are disorganized, the tissues become disorganized, and the result is impaired function. A remedy reaches the part; among the molecules of the remedy and the molecules of the protoplasmic cells chemism takes place; the toxine is antagonized, antidoted, neutralized, changed, and the metabolic process is established. This is the natural physiological process, and a healthy action results; and herein I have thought may be the secret of the action of minute doses. I will illustrate. In the chemical laboratory you have a combination of mercuric sulphate with about four-fifths its weight of chloride of sodium; and heating in a test-tube the result is mercuric chloride ($HgCl_2$), or corrosive sublimate. Repeat the experiment by using mercurous sulphate, with about a third of its weight of chloride of sodium, applying heat, and you have mercurous chloride (HgCl), calomel. Here by diminishing the amount of chlorine in one of the experiments you have quite a different result, due to quantivalence. Why, then, if the process of katabolism be chemical, may not a result be altogether different among the atoms of the molecules of the cells and of the medicine, in accordance with the law of definite proportions? If quantivalence exists among chemical radicals, why may it not exist among organic radicals?

J. F. GRIFFIN, M.D.,

The Medical Summary, Jan., 1894.

DIGESTIVE LEUCOCYTOSIS.

Many investigators, seeking after an explanation of the methods by which nutritive material is carried from the alimentary tract to the different tissues and organs of the body, have called attention to the possible importance of the white corpuscles in the assimilation of food-stuffs, citing the fact that in a well nourished carnivorous animal there is a marked production of new cells in the lymph spaces of the intestinal mucous membrane after the ingestion of food, the extent of said production being apparently dependent upon the amount of assimilable elements contained in the food. It has been further supposed that this increased production of lymph cells is followed by a corresponding increase in the passage of leucocytes into the blood, mainly on the ground that only a comparatively few of the cells could have any local action in aiding the nutrition of the intestinal walls. There has been, however, a lack of positive proof of these assumptions until Pohl,[2] in his studies on the absorption and assimilation of food-stuffs, took up the matter experimentally, and sought to obtain some positive data bearing on the question. This investigator made a careful study of the physiological variations in the content of leucocytes in dog's blood, using young animals and feeding them only one meal a day. The leucocytes were counted after Thoma's method, preliminary experiments showing that there was very little difference in the number of white cells contained in a given volume of arterial or venous blood. Thus, in the case of a fasting animal, blood taken from the jugular vein contained 16,378 white corpuscles per cubic millimeter, while blood from the carotid artery of the same side contained 15,449 white cells per cubic millimeter. Further, from a well-fed animal, whose blood was examined during digestion, similar results were obtained, *i. e.*, 27,036 from the jugular vein, and 26,866 from the carotid, thus showing that blood from the capillaries of either veins or arteries would give essentially the same results.

A single experiment, illustrative of the many reported by Pohl, may be given here, as showing the marked effect of food upon the number of white corpuscles in the circulating blood:

Time.	**Number white corpuscles per cubic millimeter blood.**
9 A. M.	8,689
9 A. M.	100 grams meat fed.
10 A. M.	16,685
11 A. M.	17,296
5 P. M.	7,256

Maximal increase, 99 per cent.

About thirty distinct experiments were tried on ten different animals, with only two or three negative results to fifty positive ones. The results, taken collectively, plainly indicate that the increase in the number of leucocytes in the circulating blood, after the ingestion of food, is very marked, the maximal increase being 146 per cent., while the average increase amounted to 78 per cent. This, as Pohl states, would indicate for the total content of blood in an animal an increase in some cases of 1,000,000,000 of leucocytes. Control experiments showed that, normally, there were only comparatively small variations in the content of leucocytes in the blood, from hour to hour, in the absence of food. It is to be further observed that this marked increase in the number of white corpuscles in the circulating blood, after the ingestion of food, usually reaches its maximum in the third hour, *viz.*: at a time when digestion would most probably have reached its height, no noticeable change being usually observed before the end of the first hour after the taking of food. Evidently, some transformation-products of the ingested food must be formed before leucocytosis becomes marked. The return to the normal number of leucocytes shows no regularity; in some cases being very gradual, in others quite rapid.

This marked action of food in increasing the number of leucocytes in the circulating blood, naturally raises the question whether all varieties of food possess this power, or whether it is limited to some one or more individual food-stuffs. In attempting to answer this question, Pohl tried a large number of experiments, the results of which afforded proof that neither water, salts, fats, carbohydrates, or even meat extracts, are able to materially affect the number of leucocytes in the blood. Proteid-containing foods, on the other hand, such as meat, Witte's peptone, and gelatin peptone quickly raise the content of white blood corpuscles to a marked degree. Somewhat noticeable was the result obtained on feeding wheat bread. This food-stuff, in spite of its fairly large content of proteid matter, failed to exert any influence on the number of leucocytes in the blood, and in conformity with this result it was found that in herbivorous animals there was an utter lack of anything approaching a digestive leucocytosis, even after protracted fasting.

In attempting to explain the cause of this increase in the number of leucocytes in the circulating blood, after the ingestion of proteid food, we are at once confronted with the possibility of this apparent increase being relative rather than absolute; as possibly due to a loss of water from the blood, incidental to the marked outpouring of digestive juices accompanying digestive proteolosis. For this view, however, there is very little support. In the first place, the blood would need to become very much thickened by loss of water to account for the increased number of leucocytes observed in the experiments. Furthermore, we know that the outpouring of watery secretions into the intestine during digestion is accompanied by an absorptive current in the opposite direction, which would tend to counteract any tendency towards concentration of the blood, and, indeed, might lead in many cases to a direct dilution of this fluid. Again, if the increase in the number of white corpuscles is to be explained in this manner, there should be a corresponding increase in the number of red blood corpuscles. As a matter of fact, Pohl's results show that those conditions of diet tending to increase the number of white blood corpuscles are without any noticeable effect on the red corpuscles. Abstraction of water can not, therefore, be the cause of the large number of leucocytes contained in the blood after a proteid diet. Much more plausible is the view that the increase is due to a more rapid transference of the corpuscles from their point of origin, *viz.*: from the intestine and from the lymph glands of the mesentery, to the blood. In other words, it seems probable that digestive proteolysis in the stomach and intestine is followed or accompanied by a rapid production of new cells in the lymph spaces surrounding the intestine. If this view is correct, there should be a much larger number of leucocytes in the blood and lymph flowing from the intestine, in an animal in full digestion, than in the arterial blood coming to the intestinal tract. That such is the case is shown by the following experiment taken from the many reported by Pohl:

DOG WEIGHING 3,330 GRAMS, IN GOOD DIGESTION.

A.M.

8.50. Blood from skin contained 8,330 white corpuscles in cubic mm.

9.10. Blood from skin contained 9,618 white corpuscles in cubic mm.

120 grams meat and 20 c. c. water fed.

10.40. Blood from the skin contained 15,092 white corpuscles in cubic mm.

10.55. Body opened, several loops made with the small intestine, and blood withdrawn from vein and artery without any great loss of blood.

WHITE CORPUSCLES IN CUBIC MM. BLOOD.

1. Intestinal loop. 11.35 A. M., venous blood contained 17,077.

   11.39 A. M., arterial blood contained 7,649.

2. Intestinal loop. 11.40 A. M., venous blood contained 15,033.

   11.40 A. M., arterial blood contained 7,061.

The facts would thus seem to warrant the assertion that, as a rule, the venous blood flowing from the intestinal tract of an animal, fed on a rich proteid diet, contains a much larger number of leucocytes than the arterial blood flowing to the intestine; although, if space permitted, we might instance certain occasional exceptions due to various causes, which, however, do not reflect against the view that there is a marked production of leucocytes in the lymph spaces surrounding the intestine, and in the lymph glands of the mesentery, as a normal accompaniment to digestive proteolysis.

Taking into account all of the circumstances attending the circulation of the blood through the abdominal organs, especially the rate of flow, and remembering the great increase in the number of leucocytes contained in the blood during the several hours attending digestive proteolysis, it is plain that a comparatively large amount of proteid matter must be transferred from the intestine to the blood in the bodies of the white corpuscles so abundantly produced in the lymph glands, etc., during proteid digestion. Indeed, Pohl calculates, on the basis of his own and the observations of others, that in the case of an animal weighing 5 kilograms, and fed with 100 grams of fresh beef, containing 20 grams of proteid matter, the entire amount of albumin required to supply the loss incidental to the normal physiological processes of the body could be absorbed into the circulation from the intestine in the form of leucocytes, assuming a digestive period of six hours. Moreover, the evidence acquired from Pohl’s experiments of the enormous production of new cells in the intestinal walls during the height of digestion renders such a theory of the transference of proteid matter from the alimentary tract to the tissues and fluids of the body quite plausible. At the end of twenty-four hours the leucocytes have fallen back to their normal number, having presumably been broken down in the blood-plasma or dissolved in the tissues and organs, thus giving up the proteid matter, of which they are composed, for the general nourishment of the body. Obviously, we can not admit that all the proteid matter of the food is absorbed in the form of leucocytes, for, as we know, some at least of the proteid food ingested is carried beyond the peptone stage, either through the action of trypsin, or through the action of the organized germs in the intestine, but we can certainly accept the views so admirably worked out by Pohl, in so far as they indicate one way in which proteid matter may pass from the intestine into the blood, after having undergone a preliminary transformation in the alimentary tract into primary albumoses and peptone. Again, if Pohl’s views are correct, we see that the proteid foods are quickly transformed into organized material in the body of the lymph cell prior to their absorption into the blood, a view which is in harmony with the long-known fact that peptone and other products of digestion can not be detected, in any quantity, at least, in the blood of the portal vein, even after the ingestion of a diet rich in proteids. As has been frequently stated in the past, the products of proteolytic action are transformed in the very act of absorption, presumably through the activity of the epithelial cells of the villi, and we may now assume, in the light of Pohl’s results, that this transformation may be due in part, at least, to the upbuilding of the ordinary products of digestion into the living protoplasm of the white corpuscles in the intestinal walls, and in this form as organized albumin circulated through the body.

Editorial in Dietetic and Hygienic Gazette, November, 1893.

Book Notices.

AN AMERICAN TEXT-BOOK OF GYNECOLOGY, MEDICAL AND SURGICAL, FOR THE USE OF STUDENTS AND PRACTITIONERS. Edited by J. M. BALDY, M. D., assisted by a corps of Nine Contributors. Cloth, 8 vo., pp. 713. 360 illustrations and 37 colored and half tone plates. Philadelphia: W. B. SAUNDERS, 1894. (Sold by Subscriptions only; Price, $6.00).

The magnificent work described above has just been received from the publisher, and inasmuch as it is at once the most modern and most complete of all works of this class prepared exclusively by American authors, it will naturally receive a warm welcome. A somewhat cursory examination shows that it is what it claims to be, namely, a practical work, written by practical men, who are, themselves well qualified by experience, and thoroughly equipped for teaching, all being teachers in high repute in different medical schools. Owing to this special feature, it is entitled to more than usual attention at the hands of American physicians; but this characteristic will give it standing abroad, and doubtless will be the means of advancing the cause of scientific treatment in the class of cases with which it has to deal. Theory and speculation have been set aside, and in their place we find reliable data not alone for the operator, but also for the general practitioner, and we bespeak for the work a favorable reception. The following extract from the prospectus will give in brief the main objects kept in view in its preparation:

In this volume all anatomical descriptions excepting what is essential to a clear understanding of the text have been omitted, illustrations being largely depended upon to elucidate this point. It will be found thoroughly practical in its teachings, and is intended, as its title implies, to be a working text-book for physicians and students. A clear line of treatment has been laid down in every case, and, although no attempt has been made to discuss mooted points, still the most important of these have been noted and explained; and the operations recommended are fully illustrated, so that the reader may have a picture of the procedure described in the text under his eye, and cannot fail to grasp the idea.

It is to be regretted that the authors' names do not appear in connection with their respective contributions, and this is about the only feature that will detract from the completeness of the work. For example, we should like to know which one of the ten is responsible for the statement found on page 90, to the effect that "*All salts of potassium in full doses are cardiac depressants.*" This is true only in part, as it is a well-recognized fact that when eliminatives are required, iodides are valuable, in fact, are demanded, and when given in what is known as medicinal doses, potassium iodide is an efficient cardiac stimulant. It is not, however, a cardiac stimulant in the sense that digitalis, strychnine and arsenic are cardiac stimulants, through their influence upon the cardiac mechanism; but rather, because potassium iodide increases protoplasmic activity, promoting the discharge of waste products, and thus lessens materially the work devolving upon the heart muscle itself. Statements of this character should always be made with caution, since they are too sweeping, and calculated to mislead those unfamiliar with the true physiological action of drugs. The book is well printed, handsomely illustrated, and reflects credit alike upon the authors and publisher.

Syllabus of Lectures on the Practice of Surgery. Arranged in conformity with the American Text-book of Surgery. By N. Senn, M. D., Ph. D., L. L. D. Cloth, 12mo., pp. 221. Philadelphia. W. B. Saunders. 1894. (Price, $2.00.)

Like all Prof. Senn's work, this is an excellent compilation, and no doubt will be highly appreciated by teachers in this department, although it will prove most acceptable to the general practitioner who desires to refresh his memory from time to time in regard to surgical affections. By reference to the list of contents, arranged alphabetically, any subject can readily be located, and all the more important points gained in the course of a few minutes. Those who do not possess the original work will be prompted to purchase it when they have an opportunity of examining this convenient arrangement.

Funny Bone.—A Book of Mirth, for Doctors, Druggists, Dentists, Medical Students and Others. Containing Funny Sayings, Jokes, Good Stories, Dialogues, Conundrums, Ludicrous Things, Ditties, etc., from a great many sources, with *Over 150 New and Original Comic Illustrations* by Dr. L. Crusius, all of which pertain to the Medical and Pharmaceutical Professions. The Funny Bone Publishing Co., 1421 Market St., St. Louis, Mo. (Price, 50 cents).

Read the preceding title and description carefully: It tells all about the nature of the book. Then take our word for it that the book is really very funny; the illustrations are unique, odd, totally unlike anything ever supplied by a comic artist, wholly original; and if you want something odd and unique, to read and to show to professional and lay friends as a curiosity, send 50 cents to the publishers and get a copy by return mail. You'll get more than 50 cents' worth of fun out of the book.

The Antikamnia Visiting List is a neat and practical account book, of convenient size, and fitted into a durable leather cover. We have received one of these lists from the Antikamnia Chemical Co., St. Louis, Mo., and have no doubt that any physician who will write to the firm can obtain the same —and make good use of it.

BOOK NOTES.

The Funk & Wagnalls Co. (18 and 20 Astor Place, New York) announce that Vol. I of their new “Standard Dictionary” was issued in December, but the first edition was not large enough to fill all orders booked before publication. A new edition is in press, and will be ready shortly. The second volume of the work will be ready in a month or two, and simultaneously the single volume edition of the entire work will be ready. Write for a prospectus: this Standard Dictionary is a wonderful book, and if you make yourself familiar with its features—as comprehensively shown in the prospectus—you are bound to secure a copy for your library. Write to the publishers at once.

E. B. Treat (5 Cooper Union, New York) has just issued two new volumes: Landis “How to Use the Forceps”, revised and enlarged by Chas. H. Bushong, M. D. (price, $1.75); and Beard’s standard work on “Nervous Exhaustion” (price, $2.75). Both books are issued in the familiar style of the publisher, and form necessary additions to available reference books on important specialties.

E. B. Treat also announces the early publication of his “International Medical Annual” for 1894 (the 14th year of publication). From the prospectus we note that the staff of editors and collaborators remains at the same high standard as heretofore, and hence the usual excellent annual review of medical progress in all branches is ensured. Write for prospectus; the book is sold at $2.75, and is cheap and indispensable.

Lea Bros. & Co. (Philada) announce the publication of the new *National Dispensatory* on January 25th. The J. B. Lippincott Co. had advertised the appearance of their new *U. S. Dispensatory* for January 15th, but some delay has apparently been caused, and the book will not appear until the middle of February. There will be greater rivalry than ever before between these two books; but each has its special features, and both are valuable—so that the careful student will do well to procure both volumes. The information in one will often amplify the other, and neither volume alone contains all the knowledge we possess—and ought to have at command—regarding drugs and materia medica.

PUBLICATIONS RECEIVED.

The Present Condition of Otology in Europe. By Lawrence Turnbull, M. D., of Philadelphia. Reprint, 1893.

The Relief of Chronic Deafness, Tinnitus Aurium and Tympanic Vertigo, by Removal of the Incus and Stapes. By Charles H. Burnett, M. D., of Philadelphia. Reprint, 1893.

Some Observations on Treating Cases of Diphtheria. By G. Benson Dunmire, A. M., M. D., of Philadelphia. Reprint, 1893.

A Case of Tumor of the Optic Thalamus. By Wharton Sinkler, M. D., of Philadelphia. Reprint, 1893.

Pathology and Treatment of Paralysis from Pott's Disease. By Wharton Sinkler, M. D., of Philadelphia. Reprint, 1893.

Syringo Myelia. By Wharton Sinkler, M. D., of Philadelphia. Reprint, 1893.

Hysterectomy: Indications and Technique. By J. M. Baldy, M. D., of Philadelphia. Reprint, 1893.

Removal of the Uterus and Its Appendages for Pelvic Inflammatory Disease. By J. M. Baldy, M. D., of Philadelphia. Reprint, 1893.

A Few Thoughts About Ophthalmometry, as to What the Javal Instrument Will Do, and What It Will Not Do. By Louis J. Lautenbach, A. M., M. D., of Philadelphia. Reprint, 1893.

Epiphora or Watery Eye. By L. Webster Fox, M. D., of Philadelphia. Reprint, 1893.

Phlyctenular Conjunctivitis, with Special Reference to the Pathology and Prophylaxis of the Disease. By Louis J. Lautenbach, A. M., M. D., of Philadelphia. Reprint, 1893.

Phthisis: A New Method of Treatment. By Henry S. Morris, M. D., of New York. Reprint. 1893.

Extract of Malt and Its Combinations. By J. J. Mulheron, M. D., of Detroit, Mich. Reprint, 1893.

The Measured Effects of Certain Therapeutic Agents, etc. By D. D. Stewart, M. D., of Philada. Reprint, 1893.

The Successful Treatment of Anæmia, with Effect Shown by Increase of Red Corpuscles and Haemoglobin. By H. P. Loomis, M. D., of New York. Reprint, 1893.

Miscellany.

METRIC EQUIVALENTS.—The metric nomenclature is coming into such common use, especially in scientific articles, that the following formulas will be found valuable:

WEIGHT EQUIVALENTS.

To convert grains into grammes multiply by	0.065
To convert grammes into grains multiply by	15.5
To convert drachms into grammes multiply by	3.9
To convert ounces (avoir.) into grammes multiply by	28.4
To convert pounds (avoir.) into grammes multiply by	453.6

MEASURE EQUIVALENTS.

To convert cubic centimeters into grains multiply by	15.5
To convert cubic centimeters into drachms multiply by	0.26
To convert cubic centimeters into ounces (avoir.) multiply by	0.036
To convert pints into cubic centimeters multiply by	473
To convert liters into ounces (avoir.) multiply by	35.3
To convert gallons into liters multiply by	3.8

FLUIDS WITH MEALS.—The arguments presented by many writers seem to prove that the moderate taking of fluids with the food at meals is not without benefit. But the importance of the thorough mastication of food before it is presented to the stomach must never be overlooked. If this is interfered with in any way by the use of liquids we must promptly prohibit their indulgence.

Fluids may be taken ad libitum during meals by those whose digestive powers will allow it; but such persons should keep in mind that the strongest stomach may be abused too far, while those whose stomachs are already unequal to a severe strain should be especially careful as to the quantity of fluid imbibed with the food.

The saliva is the best lubricator for the food while it is in the mouth, both because of its starch-digesting powers, and because its alkalinity serves to stimulate a copious flow of the acid secretion of the stomach.

Any habit, therefore, which permits the entrance of food into the stomach before it is thoroughly incorporated with saliva must be pronounced pernicious in the extreme.

If we cannot afford the time necessary for masticating our food properly and incorporating it thoroughly with saliva, it would be better to take nothing but broths and similar foods. The use of water and other liquids as lubricators is not to be tolerated.

On the other hand, if we bear in mind the whole mechanism of digestion, it will readily be seen that, in cases of weakness or want of tone on the part of the muscles of the stomach, when every part of the food cannot be properly presented to the action of the digestive juices, the introduction into the stomach of a moderate amount of water may be of no slight benefit. The mass of food will become more pliable, and so more easily operated upon by the weakened muscles.—*Youth's Companion.*

MALAKIN is a salicylated derivative of phenacetin, which occurs in small, palish-yellow crystals, insoluble in water but quite soluble in warm alcohol. The mineral acids decompose it into salicylic aldehyde and phenacetin. This also occurs in the stomach, and salicylic acid is found in the urine. Jaquet of Basle has (*Jour. de Med. de Paris*) found it of value in rheumatics in whom salicylic acid produces untoward effects. According to him, it has a mild, efficient action similar to that of nascent salicylic acid. No untoward effects were observed, but the results were prompt. It is given in 15-grain doses four or six times daily. As an anti-neuralgic and antipyretic it is inferior to phenacetin.—*Med. Standard.*

PROTECTION AGAINST DIPHTHERIA.—The Board of Health has announced a new measure looking to the control and diminution of diphtheria, and circulars were sent to practising physicians giving the grounds for the step decided upon, and the reasons why it was deemed expedient.

The proposition is to supplement the primary bacteriological examination now made at the beginning of any individual case of the disease, by other cultures repeated during its course and during convalescence. It is hoped in this way to make sure that apparent recovery, and the disappearance of all false membrane is followed by the extermination of all the Loeffler bacilli from the throat. The circular is written by Dr. Hermann M. Biggs, chief inspector of pathology, bacteriology, and disinfection, and is signed by President Wilson with the approval of the board. It is explained that 405 cases of true diphtheria have been subjected to repeated examinations at intervals of three or four days during illness and until the disappearance of the bacilli. It was found that in 160 cases the bacilli persisted after the complete separation of the false membrane, or, in other words, after the individual had recovered. Of these 160 cases, 103 showed the germ for seven days, thirty-four for twelve days, sixteen for fifteen days, four for three weeks, and in three for five weeks, after the exudation had completely disappeared from the upper air-passages. The circular infers, thence, that "these results show that in a considerable proportion of cases persons who have had diphtheria continue to carry the germs of the disease in their throats for many days after all signs and symptoms of the disease have disappeared."

These experiments have led the Health Department to adopt the rule that no person who has suffered from diphtheria shall be considered free from contagion until it has been shown by bacteriological examination, made after the disappearance of the membrane from the throat, that the throat secretions no longer contain the diphtheria bacilli, and that until such examinations have shown such absence all cases in boarding-houses, hotels, and tenement-houses must remain isolated and under observation. Disinfection of the premises, therefore, will not be performed by the department until examination has shown the absence of the organisms.

Secondary cultures, as in the case of primary cultures, may be made by the attending physician, if he so desires; otherwise they will be made by the inspector of the district in which the case occurs. This applies only to cases occurring in boarding-houses, hotels, and tenement-houses—not to those in private houses.—*N. Y. Evening Post*, January 8, 1894.

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1. The source is not quite so accurately known; hippuric acid is eliminated with the urine in human subjects as well—after the administration of benzoic acid. For an interesting compilation on

this point, see Wood's Therapeutics, page 642.

2. Archiv für Experimentelle Pathol. und Pharm., Band 25, p. 31.

TRANSCRIBER'S NOTES

1. Silently corrected typographical errors and variations in spelling.
2. Retained anachronistic, non-standard, and uncertain spellings as printed.
3. Footnotes have been re-indexed using numbers and collected together at the end of the last chapter.

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